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How Do Short-Term Incentives Affect Long-Term Productivity?

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1 Overview

1.1 Question

The paper studies a specific and economically important form of corporate short-termism:

When firms face incentives to avoid missing earnings-per-share (EPS) targets, do they engage in EPS-motivated share buybacks, and do these actions harm long-run productivity?

1.2 Core empirical finding

Using U.S. Census plant-level manufacturing data (CMF/ASM) linked to Compustat/CRSP/IBES, the paper finds that short-term incentives to engage in *EPS-motivated buybacks* lead to:

- **lower productivity** (TFP) at both the firm and plant level in the subsequent three years;
- **cuts to investment and employment**, including cuts to **productivity-augmenting technologies** (machinery and information technology investments);
- **inefficient within-firm allocation**: when resources shrink, firms cut across plants in a way that is not well-targeted toward ex ante unproductive plants.

1.3 Mechanisms emphasized by the paper

The productivity decline is amplified when firms face frictions that impede efficient reallocation:

- (i) **labor frictions** (union power; right-to-work exposure),
- (ii) **financial constraints** (Whited–Wu index; lack of bond ratings),
- (iii) **agency conflicts** (short-horizon investors; CEOs whose pay depends explicitly on EPS),
- (iv) **convex adjustment costs** that make concentrated cuts more costly.

2 Data, sample, and key objects

2.1 Plant-level Census data and sample

- Plant-level inputs/outputs from the Census of Manufactures (CMF) and Annual Survey of Manufactures (ASM).
- Linked to Compustat/CRSP/IBES to construct EPS surprises and buyback-related objects.
- Final sample (manufacturing): **3,300 firm-year** focal observations covering about **35,000 plant-years**, 1988–2013 (Table 1).

2.2 Why plant-level data matter

Compared with accounting-only firm-level data, plant-level census microdata allow:

- more precise measurement of TFP using detailed input and output components;
- decomposition of firm outcomes into *plant-level* productivity changes and *within-firm reallocation* across plants;
- observation of investment composition (e.g., machinery and IT/computers) that is closely tied to productivity growth.

3 Economic setup: why buybacks can move EPS

3.1 Accounting mechanics (model-free but essential)

EPS is an accounting ratio:

$$EPS = \frac{\text{Earnings}}{\text{Shares outstanding}}.$$

A share repurchase reduces shares outstanding, mechanically increasing EPS. But using cash for buybacks also reduces earnings via *foregone interest* (or financing costs). Thus the net effect depends on both numerator and denominator changes.

3.2 Pre-repurchase EPS (constructed counterfactual)

The paper constructs a counterfactual EPS that reflects what EPS would have been *absent repurchases*. Let:

- E_{ft} = reported earnings,
- S_{ft} = end-of-quarter shares outstanding,
- B_{ft} = repurchase spending,
- P_{ft} = average share price (used to infer shares repurchased),
- $\Delta S_{ft} := B_{ft}/P_{ft}$ = estimated shares repurchased,
- I_{ft} = after-tax interest that would have been earned on the repurchase funds if invested in a 3-month T-bill.

Then the repurchase-adjusted (“pre-repurchase”) EPS is:

$$EPS_{ft}^{pre} = \frac{E_{ft} + I_{ft}}{S_{ft} + \Delta S_{ft}}. \tag{1}$$

This is the key object for constructing the *running variable* in the regression discontinuity design below.

4 Identification: fuzzy regression discontinuity around a pre-EPS threshold

4.1 Running variable: pre-repurchase EPS surprise

Let F_{ft} be the median end-of-quarter analyst EPS forecast. Define the *pre-repurchase EPS surprise* (scaled by price) as:

$$Surp_{ft}^{pre} = \frac{EPS_{ft}^{pre} - F_{ft}}{P_{ft}}. \quad (2)$$

Intuition:

- If $Surp_{ft}^{pre} < 0$, the firm would *just miss* the forecast absent buybacks.
- If $Surp_{ft}^{pre} \geq 0$, the firm would *meet/beat* absent buybacks.

4.2 Treatment indicator: incentive to do EPS-motivated repurchases

Define:

$$D_{ft} := \mathbf{1}\{Surp_{ft}^{pre} < 0\}.$$

This is *not* an indicator of whether the firm actually repurchased. It captures whether the firm has a *discontinuous incentive* to repurchase to avoid missing the EPS benchmark.

4.3 Why the design is “fuzzy”

The probability/amount of buybacks increases discontinuously when $Surp_{ft}^{pre}$ crosses 0 from above to below, but buybacks are not deterministic at the threshold. Hence, D_{ft} shifts the intensity of buybacks *on average* rather than assigning treatment perfectly.

4.4 Outcome construction: long-run changes relative to pre-period

For an outcome Y , the main dependent variable is defined as:

$$\Delta Y_{i,t} = \underbrace{\frac{1}{3} \sum_{k=1}^3 Y_{i,t+k}}_{\text{post: 3-year average}} - \underbrace{Y_{i,t-1}}_{\text{pre: lag}}. \quad (3)$$

For dynamics (Table 6), the paper also uses $\Delta Y_{i,t}^{(k)} = Y_{i,t+k} - Y_{i,t-1}$.

4.5 Baseline reduced-form fuzzy RD regression (Equation (1))

Within a narrow bandwidth around the threshold (baseline window $[-0.003, +0.003]$), the paper estimates:

$$\Delta Y_{i,t} = \beta_1 D_{i,t} + \beta_2 Surp_{i,t}^{pre} + \beta_3 (Surp_{i,t}^{pre} \times D_{i,t}) + \beta_4' X_{i,t} + \eta_t + \varepsilon_{i,t}, \quad (4)$$

where:

- i indexes firms (or plants in plant-level regressions),

- $X_{i,t}$ includes controls (e.g., plant size and age in plant regressions),
- η_t are year fixed effects in firm regressions; in plant regressions, the paper includes industry $\times$ year and state $\times$ year fixed effects plus plant controls,
- standard errors are clustered at the firm level.

Identification assumption (local smoothness). Absent the discontinuous incentive to repurchase at $Surp^{pre} = 0$, potential outcomes evolve smoothly in $Surp^{pre}$. Conditional on the local polynomial in $Surp^{pre}$, there is no other discontinuous shift at 0 that drives future productivity.

5 Productivity measurement and resource-allocation variables

5.1 Plant-level TFP: index approach (Equation (2))

Plant-level total factor productivity (TFP) is measured as (Syverson’s index approach):

$$TFP_{p,t} = \ln Q_{p,t} - \alpha_k \ln K_{p,t} - \alpha_\ell \ln L_{p,t} - \alpha_m \ln M_{p,t}, \quad (5)$$

where Q is real output, K capital stock, L labor input, M materials, and $(\alpha_k, \alpha_\ell, \alpha_m)$ are factor elasticities (cost shares). Output is deflated using NBER-CES industry price indices.

5.2 Firm-level TFP aggregation

Firm-level $TFP_{f,t}$ is constructed as a **capital-weighted average** of its plants’ TFP (Giroud & Mueller 2015; Schoar 2002). The main firm-level dependent variable is the change in firm-level TFP as in (3).

5.3 Resource allocation outcomes

Key outcomes include:

- **Employment** (payroll / employment expenditures), and variants (log employees; symmetric growth),
- **Investment** (capital expenditures), including subcategories:
 - machinery investment,
 - computer (IT) investment,
- **Separation** indicator: whether the plant is sold or closed in the post window (extensive margin).

5.4 Within-firm heterogeneity: productive vs. unproductive plants

For within-firm allocation tests:

- $Productive_{p,t-1} = 1$ if plant p has above-median *within-firm* TFP at $t - 1$,
- $Unproductive_{p,t-1} = 1$ if below-median *within-firm* TFP at $t - 1$.

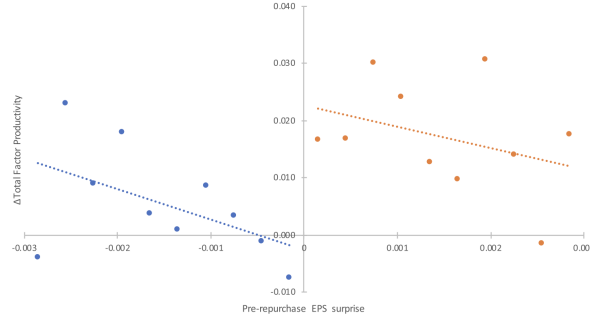


Figure 1
Negative pre-repurchase EPS surprises and change in productivity among manufacturing plants

Table 2
Short-term incentives and productivity

A. Firm-level analysis

Dependent variable:	ΔTFP [1]
Negative pre-repurchase EPS surprise	-0.025** (0.013)
Linear control in pre-repurchase EPS surprise	Y
Year fixed effects	Y
Rounded N	3,300
R^2	0.012

B. Plant-level analysis

Dependent variable:	ΔTFP [1]
Negative Pre-Repurchase EPS Surprise	-0.019*** (0.009)
Linear control in pre-repurchase EPS surprise	Y
Plant controls	Y
Industry $\times$ year fixed effects	Y
State $\times$ year fixed effects	Y
Rounded N	35,000
R^2	0.085

6 Main results: productivity falls after EPS-motivated incentives

6.1 Figure 1: visual evidence of a discontinuity

Figure 1 plots binned averages of future TFP changes against $Surp^{pre}$. The fitted lines on each side show a visible downward jump at $Surp^{pre} = 0$: plants associated with negative pre-repurchase surprises experience lower subsequent TFP growth.

6.2 Table 2: baseline reduced-form estimates

Table 2 implements (4) with ΔTFP as the outcome:

- **Firm-level:** D_{ft} has a negative coefficient (about -0.025), statistically significant.
- **Plant-level:** the negative effect is also present and statistically stronger (about -0.019), indicating the productivity drop is not merely an aggregation artifact.

Table 3
Short-term incentives and plant-level resource allocation

A. Plant-level analysis

Dependent variable:	Δ Employment [1]	Δ Investment [2]	Separation [3]
<i>Negative pre-repurchase EPS surprise</i>	−0.022*** (0.006)	−0.006* (0.003)	−0.005 (0.012)
Linear control in pre-repurchase EPS surprise	Y	Y	Y
Plant controls	Y	Y	Y
Industry $\times$ year fixed effects	Y	Y	Y
State $\times$ year fixed effects	Y	Y	Y
Rounded N	35,000	35,000	43,500
R^2	0.111	0.064	0.108

B. Investment types

Dependent variable:	Δ Machinery investment [1]	Δ Computer investment [2]
<i>Negative pre-repurchase EPS surprise</i>	−0.004** (0.002)	−0.001* (0.001)
Linear control in pre-repurchase EPS surprise	Y	Y
Plant controls	Y	Y
Industry $\times$ year fixed effects	Y	Y
State $\times$ year fixed effects	Y	Y
Rounded N	35,000	14,000
R^2	0.066	0.201

C. Employment types

Dependent variable:	Δ Production employment [1]	Δ Non-production employment [2]
<i>Negative pre-repurchase EPS surprise</i>	−0.013*** (0.004)	−0.008** (0.004)
Linear control in pre-repurchase EPS surprise	Y	Y
Plant controls	Y	Y
Industry $\times$ year fixed effects	Y	Y
State $\times$ year fixed effects	Y	Y
Rounded N	35,000	35,000
R^2	0.113	0.082

Interpretation. The estimates imply that short-term incentives that raise buybacks near EPS thresholds are associated with *real declines in production efficiency* (TFP), not merely with changes in scale.

7 Resource-allocation channel: investment and employment cuts

7.1 Table 3: overall plant-level cuts and their composition

Table 3 estimates (4) at the plant level with resource-allocation outcomes:

- **Employment falls** (column 1, panel A): negative surprise is associated with lower payroll over the next three years.
- **Investment falls** (column 2, panel A): firms cut capital expenditures.
- **No strong extensive-margin response** (separation; column 3, panel A): the reduction occurs mostly within existing plants (intensive margin).

- **Technology-embodied capital is cut** (panel B): machinery and computer (IT) investment both decline, consistent with reduced adoption of productivity-augmenting technologies.
- **Both worker types are affected** (panel C): reductions in both production and non-production employment.

Interpretation. These patterns explain why TFP falls: the cuts are not obviously “trimming low-NPV fat” only. They include investment types typically linked to technological upgrading (machines and IT) that support future productivity.

8 Within-firm allocation: why productivity can fall even if firms cut scale

A firm can shrink without losing productivity *if* it targets cuts to its least productive units. The plant data allow direct tests of whether cuts are concentrated in ex ante unproductive plants.

8.1 Table 4: are cuts targeted to unproductive plants? (answer: mostly no)

Table 4 interacts the RD incentive D_{ft} with plant productivity type (productive vs unproductive at $t - 1$). Main reading:

- **Employment and investment cuts occur in both productive and unproductive plants** and are generally **statistically similar in magnitude** (panel A and panel B: p-values for differences are large).
- **Separation is not the main margin:** there is little evidence that firms systematically close/sell specific plants to improve efficiency (panel A, column 3).
- **A particularly inefficient pattern appears in some labor components:** in panel C for non-production employment, the cut is larger in productive plants (difference significant at conventional levels).

Interpretation. The firm does *not* respond to the resource shortfall by disproportionately trimming low-productivity plants. Instead, it cuts in a relatively broad-brush way across plants, consistent with frictions in reallocation.

8.2 Table 5: which plants’ productivity declines?

Table 5 estimates the plant-level change in TFP by plant type. Key point:

- The negative TFP response is **not concentrated in unproductive plants**; the decline is present (and statistically stronger) for **productive** plants.

Intuition. If productive plants are precisely the units where technology adoption and efficient input use matter most, indiscriminate investment cuts can reduce their productivity and produce a larger firm-level TFP loss.

Table 4
Plant-level resource allocation effects: High- vs. low-productivity plants

A. Employment, investment, and separation

Dependent variable:	Δ Employment [1]	Δ Investment [2]	Separation [3]
$Unproductive_{t-1}$	-0.014* (0.006)	-0.006** (0.003)	-0.008 (0.007)
Negative pre-repurchase EPS surprise $\times$ $Productive_{t-1}$	-0.024*** (0.007)	-0.004 (0.004)	-0.013 (0.012)
Negative pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	-0.019*** (0.007)	-0.007* (0.004)	0.003 (0.014)
Linear control in pre-repurchase EPS surprise	Y	Y	Y
Linear control in pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	Y	Y	Y
Plant controls	Y	Y	Y
Industry $\times$ year fixed effects	Y	Y	Y
State $\times$ year fixed effects	Y	Y	Y
Rounded N	35,000	35,000	43,500
R^2	0.112	0.065	0.108
F-stat	0.31	0.25	1.88
p -value of difference between prod. and unprod. interaction terms	0.58	0.61	0.17

B. Investment types

Dependent variable:	Δ Machinery investment [1]	Δ Computer investment [2]
$Unproductive_{t-1}$	-0.006*** (0.002)	-0.000 (0.001)
Negative pre-repurchase EPS surprise $\times$ $Productive_{t-1}$	-0.004 (0.003)	-0.001** (0.001)
Negative pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	-0.005* (0.003)	-0.001 (0.001)
Linear control in pre-repurchase EPS surprise	Y	Y
Linear control in pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	Y	Y
Plant controls	Y	Y
Industry $\times$ year fixed effects	Y	Y
State $\times$ year fixed effects	Y	Y
Rounded N	35,000	14,000
R^2	0.067	0.202
F-stat	0.16	1.72
p -value of difference between prod. and unprod. interaction terms	0.69	0.19

C. Employment types

Dependent variable:	Δ Production employment [1]	Δ Non-production employment [2]
$Unproductive_{t-1}$	-0.001 (0.004)	-0.013*** (0.004)
Negative pre-repurchase EPS surprise $\times$ $Productive_{t-1}$	-0.011** (0.005)	-0.013*** (0.004)
Negative pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	-0.016*** (0.005)	-0.003 (0.005)
Linear control in pre-repurchase EPS surprise	Y	Y
Linear control in pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	Y	Y
Plant controls	Y	Y
Industry $\times$ year fixed effects	Y	Y
State $\times$ year fixed effects	Y	Y
Rounded N	35,000	35,000
R^2	0.113	0.084
F-stat	0.77	4.23
p -value of difference between prod. and unprod. interaction terms	0.38	0.04

Table 5
Changes to plant-level productivity: ex ante high- vs. low-productivity plants

Dependent variable:	ΔTFP [1]
$Unproductive_{t-1}$	-0.175*** (0.013)
Negative pre-repurchase EPS surprise $\times$ $Productive_{t-1}$	-0.023** (0.010)
Negative pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	-0.014 (0.015)
Linear control in pre-repurchase EPS surprise	Y
Linear control in pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	Y
Plant controls	Y
Industry $\times$ year fixed effects	Y
State $\times$ year fixed effects	Y
Rounded N	35,000
R^2	0.149
F-stat	0.51
p -value of difference between prod. and unprod. interaction terms	0.48

Table 6
Dynamics of effects

Dependent variable:	$\Delta Investment$			$\Delta Employment$			ΔTFP		
	(-1,+1) [1]	(-1,+2) [2]	(-1,+3) [3]	(-1,+1) [4]	(-1,+2) [5]	(-1,+3) [6]	(-1,+1) [7]	(-1,+2) [8]	(-1,+3) [9]
Negative pre-repurchase EPS surprise	-0.013** (0.012)	-0.015** (0.014)	-0.019** (0.016)	-0.021 (0.017)	-0.044** (0.021)	-0.059** (0.027)	-0.025** (0.006)	-0.024** (0.007)	-0.027** (0.007)
Linear control in pre-repurchase EPS surprise	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Rounded N	3,300	3,300	3,300	3,300	3,300	3,300	3,300	3,300	3,300
R^2	0.013	0.016	0.011	0.034	0.032	0.028	0.028	0.024	0.024

9 Dynamics: when do the effects show up? (Table 6)

Table 6 measures year-by-year changes:

- **Investment cuts appear quickly** and persist through year $t + 3$.
- **Employment declines are slower to materialize:** the point estimate grows in magnitude and becomes significant at longer horizons.
- **TFP declines persistently:** negative and significant at $t + 1$, $t + 2$, and $t + 3$.

Interpretation. The timing is consistent with the idea that investment cuts (especially in technology) can have delayed but persistent productivity consequences.

10 Mechanisms: why do firms fail to reallocate efficiently? (Tables 7–11)

The paper argues that firms would ideally concentrate cuts in low-productivity plants. Why do they not? The central answer is **frictions that constrain reallocation**.

10.1 Labor frictions (Table 7): union power and right-to-work exposure

The paper uses right-to-work (RTW) laws as a proxy for weaker union bargaining power. Two firm-level exposure measures are used:

Table 7
Potential mechanisms: State-level union power

A. Firm-level productivity growth

Firm splits:	Average effect [1]	% plants in RTW states		% production hours in RTW states	
		Above med. [2]	Below med. [3]	Above med. [4]	Below med. [5]
Dependent variable: ΔTFP					
Negative pre-repurchase EPS surprise	-0.025** (0.013)	-0.005 (0.016)	-0.047** (0.020)	-0.009 (0.016)	-0.045** (0.021)
Linear control in pre-repurchase EPS surprise	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y
Rounded N	3,300	1,700	1,700	1,700	1,700
R^2	0.012	0.022	0.018	0.028	0.016
p -value of difference		0.09		0.10	

B. Firm-level productivity growth and employment cuts

Firm splits:	Average effect [1]	% plants in RTW states		% production hours in RTW states	
		Above med. [2]	Below med. [3]	Above med. [4]	Below med. [5]
Dependent variable: ΔTFP					
Negative pre-repurchase EPS surprise	-0.015 (0.021)	-0.041 (0.025)	-0.003 (0.020)	-0.035 (0.024)	-0.032 (0.025)
Employment cut	-0.021 (0.019)	-0.021 (0.023)	-0.013 (0.021)	-0.032 (0.025)	-0.040* (0.023)
Negative pre-repurchase EPS surprise $\times$ Employment cut	0.044 (0.033)	-0.075** (0.037)	0.009 (0.034)	-0.040* (0.023)	
Linear control in pre-repurchase EPS surprise	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y
Rounded N	1,700	1,700	1,700	1,700	1,700
R^2	0.023	0.021	0.028	0.017	

C. Plant-level outcomes

Union power:	RTW state			Non-RTW state		
	$\Delta Employment$ [1]	$\Delta Investment$ [2]	Separation [3]	$\Delta Employment$ [4]	$\Delta Investment$ [5]	Separation [6]
Dependent variable:						
$Unproductive_{t-1}$	-0.005 (0.009)	-0.002 (0.004)	-0.006 (0.010)	-0.022*** (0.008)	-0.009*** (0.003)	-0.010 (0.008)
Negative pre-repurchase EPS surprise $\times$ $Productive_{t-1}$	-0.015 (0.011)	0.002 (0.006)	-0.000 (0.017)	-0.027*** (0.009)	-0.007 (0.005)	-0.021 (0.013)
Negative pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	-0.002 (0.011)	-0.002 (0.005)	0.033** (0.017)	-0.029*** (0.008)	-0.010** (0.005)	-0.017 (0.016)
Linear control in pre-repurchase EPS surprise	Y	Y	Y	Y	Y	Y
Linear control in pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	Y	Y	Y	Y	Y	Y
Plant controls	Y	Y	Y	Y	Y	Y
Industry $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
State $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
Rounded N	14,000	14,000	17,500	21,000	21,000	26,000
R^2	0.139	0.088	0.154	0.116	0.069	0.160

- fraction of plants in RTW states,
- fraction of production hours in RTW states.

Panel A (firm-level TFP). The TFP decline is concentrated among firms with **lower RTW exposure** (i.e., stronger union power environments).

Panel B (TFP and employment cuts). The TFP decline is strongest when firms in high-union-power environments actually reduce the number of employees (an “employment cut” state), consistent with labor reallocation constraints binding most when layoffs occur.

Panel C (plant-level allocation). Plants in RTW states show comparatively more *selective* adjustments (including evidence consistent with separating unproductive plants), whereas plants in non-RTW states display more indiscriminate cuts across productive/unproductive plants.

10.2 Financial constraints (Table 8): constrained firms drive the effects

Two measures:

- Whited–Wu (2006) index split (above-median = constrained),
- bond rating split (no rating = constrained).

Table 8						
Potential mechanisms: Financial constraints						
A. Firm-level productivity growth						
Dependent variable: ΔTFP						
	Whited and Wu (2006)		Bond rating			
Constrained firm?	Yes [1]	No [2]	Yes [3]	No [4]		
Negative pre-repurchase EPS surprise	-0.056** (0.022)	-0.002 (0.017)	-0.051*** (0.018)		-0.002 (0.017)	
Linear control in pre-repurchase EPS surprise	Y	Y	Y		Y	
Year fixed effects	Y	Y	Y		Y	
Rounded N	1,700	1,700	1,500		1,800	
R^2	0.027	0.028	0.022		0.018	
F-stat	3.62		3.23			
p-value of difference	0.06		0.07			
B. Plant-level outcomes						
Financial constraints measure:						
	Whited and Wu (2006): Constrained			Whited and Wu (2006): Unconstrained		
Dependent variable:	$\Delta Employment$ [1]	$\Delta Investment$ [2]	Separation [3]	$\Delta Employment$ [4]	$\Delta Investment$ [5]	Separation [6]
$Unproductive_{t-1}$	-0.026* (0.014)	-0.013* (0.007)	-0.010 (0.017)	-0.013** (0.007)	-0.005* (0.003)	-0.004 (0.010)
Negative pre-repurchase EPS surprise $\times$ $Productive_{t-1}$	-0.063** (0.026)	-0.014* (0.007)	-0.076** (0.032)	-0.013 (0.009)	-0.003 (0.004)	-0.019 (0.018)
Negative pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	-0.070** (0.032)	-0.013* (0.007)	-0.047 (0.040)	-0.009 (0.007)	-0.006 (0.014)	-0.012 (0.020)
Linear control in pre-repurchase EPS surprise	Y	Y	Y	Y	Y	Y
Linear control in pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	Y	Y	Y	Y	Y	Y
Plant controls	Y	Y	Y	Y	Y	Y
Industry $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
State $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
Rounded N	6,300	6,300	7,900	29,000	29,000	35,500
R^2	0.272	0.237	0.278	0.118	0.072	0.146
F-stat	0.04	0.00	0.60	0.25	0.64	0.20
p-value of difference between prod. and unprod. interaction terms	0.83	0.95	0.44	0.62	0.51	0.66
C. Plant-level outcomes						
Financial constraints measure:						
	Bond rating: Constrained			Bond rating: Unconstrained		
Dependent variable:	$\Delta Employment$ [1]	$\Delta Investment$ [2]	Separation [3]	$\Delta Employment$ [4]	$\Delta Investment$ [5]	Separation [6]
$Unproductive_{t-1}$	-0.023* (0.013)	-0.013* (0.007)	-0.010 (0.015)	-0.014** (0.007)	-0.003 (0.003)	-0.003 (0.011)
Negative pre-repurchase EPS surprise $\times$ $Productive_{t-1}$	-0.040* (0.023)	-0.011* (0.007)	-0.011 (0.035)	-0.016 (0.017)	-0.003 (0.004)	-0.028 (0.019)
Negative pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	-0.058** (0.028)	-0.020* (0.011)	0.017 (0.035)	-0.010 (0.007)	-0.007 (0.014)	-0.019 (0.021)
Linear control in pre-repurchase EPS surprise	Y	Y	Y	Y	Y	Y
Linear control in pre-repurchase EPS surprise $\times$ $Unproductive_{t-1}$	Y	Y	Y	Y	Y	Y
Plant controls	Y	Y	Y	Y	Y	Y
Industry $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
State $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
Rounded N	7,800	7,800	9,600	27,500	27,500	34,000
R^2	0.246	0.199	0.259	0.117	0.072	0.150
F-stat	0.47	0.37	0.99	0.51	0.80	0.29
p-value of difference between prod. and unprod. interaction terms	0.49	0.54	0.32	0.47	0.37	0.59

Table 9
Potential mechanisms: Investor horizon

A. Firm-level productivity growth

Dependent variable: ΔTFP	Long-term investors		
	Average effect [1]	Below median [2]	Above median [3]
<i>Negative pre-repurchase EPS surprise</i>	-0.025** (0.013)	-0.045** (0.019)	-0.006 (0.017)
Linear control in pre-repurchase EPS surprise	Y	Y	Y
Year fixed effects	Y	Y	Y
Rounded N	3,300	1,700	1,700
R^2	0.012	0.022	0.013
p -value of difference		0.10	

B. Plant-level outcomes

Long-term investors:	Below median			Above median		
	$\Delta Employment$ [1]	$\Delta Investment$ [2]	Separation [3]	$\Delta Employment$ [4]	$\Delta Investment$ [5]	Separation [6]
<i>Unproductive_{$t-1$}</i>	-0.016** (0.008)	-0.010*** (0.003)	-0.000 (0.0010)	-0.009 (0.009)	0.002 (0.004)	-0.005 (0.017)
<i>Negative pre-repurchase EPS surprise $\times$ Productive_{$t-1$}</i>	-0.031*** (0.009)	-0.007* (0.004)	-0.021 (0.020)	-0.007 (0.013)	0.002 (0.008)	-0.013 (0.024)
<i>Negative pre-repurchase EPS surprise $\times$ Unproductive_{$t-1$}</i>	-0.022** (0.010)	-0.007* (0.004)	-0.013 (0.021)	-0.019 (0.013)	-0.011* (0.007)	-0.006 (0.029)
Linear control in pre-repurchase EPS surprise	Y	Y	Y	Y	Y	Y
Linear control in pre-repurchase EPS surprise $\times$ Unproductive _{$t-1$}	Y	Y	Y	Y	Y	Y
Plant controls	Y	Y	Y	Y	Y	Y
Industry $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
State $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
Rounded N	19,000	19,000	24,000	16,000	16,000	19,000
R^2	0.135	0.088	0.174	0.188	0.129	0.204
F-stat	0.84	0.00	0.30	0.71	1.64	0.08
p -value of difference between prod. and unprod. interaction terms	0.36	0.98	0.58	0.40	0.20	0.78

Panel A (firm-level TFP). TFP declines significantly only among **financially constrained** firms; unconstrained firms show near-zero effects.

Panels B–C (plant-level outcomes). Constrained firms cut employment and investment strongly and across both productive and unproductive plants. Unconstrained firms exhibit weaker and less systematic cut patterns.

Interpretation. If unconstrained firms can fund buybacks without cutting productivity-critical inputs, then the productivity loss is an interaction of buyback incentives with binding financing constraints.

10.3 Agency conflicts: investor horizon (Table 9)

Using Bushee’s classification of 13F investors:

- Firms with more **short-horizon** investors (below-median long-term ownership) experience large negative TFP effects.
- Firms with more **long-horizon** investors show little or no TFP deterioration.

Plant-level allocation (Table 9, panel B). Short-horizon firms cut across both plant types (inefficient). Long-horizon firms’ cuts are more concentrated in unproductive plants (more efficient), consistent with the absence of TFP decline.

10.4 Agency conflicts: CEO pay depends on EPS (Table 10)

Using compensation plan disclosures (ISS Incentive Lab):

- The productivity decline is concentrated in firms where the CEO’s pay **explicitly depends on EPS**.

Table 10
Potential mechanisms: Contracting frictions

A. Firm-level productivity growth

Dependent variable: ΔTFP

CEO's pay depends on EPS?

Firm splits:	Average effect [1]	Yes [2]	No [3]
Negative pre-repurchase EPS surprise	-0.038* (0.023)	-0.094*** (0.035)	-0.006 (0.031)
Linear control in pre-repurchase EPS surprise	Y	Y	Y
Year fixed effects	Y	Y	Y
Rounded N	1,100	350	700
R^2	0.025	0.072	0.019
F-stat		2.83	
p-value of difference		0.09	

B. Plant-level outcomes

Contract status:

Dependent variable:

	CEO's pay depends on EPS: Yes			CEO's pay depends on EPS: No		
	$\Delta Employment$ [1]	$\Delta Investment$ [2]	Separation [3]	$\Delta Employment$ [4]	$\Delta Investment$ [5]	Separation [6]
Unproductive $_{t-1}$	-0.016 (0.011)	-0.001 (0.006)	-0.019 (0.018)	-0.025* (0.014)	-0.083* (0.008)	0.010 (0.027)
Negative pre-repurchase EPS surprise $\times$ Productive $_{t-1}$	-0.030** (0.014)	-0.014* (0.008)	-0.026 (0.028)	-0.017 (0.016)	0.005 (0.008)	0.014 (0.027)
Negative pre-repurchase EPS surprise $\times$ Unproductive $_{t-1}$	0.007 (0.012)	-0.017* (0.009)	0.002 (0.035)	-0.014 (0.017)	-0.005 (0.009)	-0.002 (0.024)
Linear control in pre-repurchase EPS surprise	Y	Y	Y	Y	Y	Y
Linear control in pre-repurchase EPS surprise $\times$ Unproductive $_{t-1}$	Y	Y	Y	Y	Y	Y
Plant controls	Y	Y	Y	Y	Y	Y
Industry $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
State $\times$ year fixed effects	Y	Y	Y	Y	Y	Y
Rounded N	5,600	5,600	6,900	8,200	8,200	10,500
R^2	0.160	0.143	0.490	0.175	0.106	0.405
F-stat	5.67	0.07	0.57	0.04	0.83	0.70
p-value of difference between prod. and unprod. interaction terms	0.02	0.79	0.45	0.85	0.36	0.40

Table 11
Potential mechanisms: Convex adjustment costs

Dependent variable: ΔTFP

Within-firm standard deviation of TFP

Firm splits:	[1]	Low [2]	High [3]
Negative pre-repurchase EPS surprise	0.003 (0.031)	-0.007 (0.032)	0.014 (0.078)
Investment cut HHI	0.026 (0.021)	0.022 (0.026)	0.032 (0.033)
Negative pre-repurchase EPS surprise $\times$ Investment cut HHI	-0.087** (0.033)	-0.066* (0.039)	-0.126** (0.064)
Linear control in pre-repurchase EPS surprise	Y	Y	Y
Year fixed effects	Y	Y	Y
Rounded N	3,300	1,700	1,700
R^2	0.015	0.021	0.024

- These firms cut employment and investment across plants, including productive plants, suggesting stronger short-term distortion.

Interpretation. An explicit EPS contract maps the accounting threshold into a real incentive to undertake potentially value-destroying actions.

10.5 Convex adjustment costs and “cut concentration” (Table 11)

The paper constructs an *investment-cut concentration* index analogous to HHI:

$$HHI_{f,t}^{cut} = \sum_{p \in f} s_{p,f,t}^2, \quad s_{p,f,t} = \frac{\text{investment cut at } p}{\sum_{q \in f} \text{investment cut at } q}.$$

Large values indicate cuts concentrated in a few plants; small values indicate cuts spread broadly.

Table 11 augments (4) with the interaction $D_{f,t} \times HHI_{f,t}^{cut}$. Key reading:

- The interaction is **negative**: when cuts are more concentrated, the negative TFP effect is stronger. This is consistent with **convex adjustment costs**—large concentrated investment adjustments are more costly and harm productivity.
- The interaction remains (and is stronger) among firms with high within-firm TFP dispersion, suggesting that even when firms have more scope to target low-productivity units, adjustment frictions can dominate.

11 Synthesis

11.1 A causal narrative consistent with the evidence

Putting all results together suggests the following chain:

1. Near the EPS threshold, firms face a **discontinuous incentive** to use share repurchases to avoid a miss.
2. Those incentives are associated with subsequent **cuts in investment and employment**, including technology-embodied capital.
3. Because firms face **frictions in reallocation** (labor constraints, financing constraints, agency problems, adjustment costs), they fail to concentrate cuts in low-productivity units.
4. As a result, productivity declines at productive plants and, therefore, at the firm level.

11.2 Summary: what the paper concludes

This paper provides causal evidence that short-term incentives tied to meeting quarterly EPS benchmarks can have economically meaningful long-run real costs. The authors exploit a fuzzy regression discontinuity design around a *pre-repurchase* EPS surprise threshold. Firms that would *just miss* the earnings forecast absent repurchases face a discontinuous incentive to repurchase shares to avoid an EPS miss. Around this threshold, the paper finds that:

- (i) **EPS-avoidance incentives increase repurchases.** The probability and intensity of buybacks rise discretely when firms are marginally below the pre-repurchase EPS threshold, consistent with repurchases being used to manage EPS outcomes.
- (ii) **Long-run productivity declines after EPS-motivated incentives.** Firms (and their plants) exposed to the discontinuous EPS-miss incentive experience lower subsequent productivity growth measured by total factor productivity (TFP), with effects persisting over multiple years.
- (iii) **The productivity decline is accompanied by cuts to real inputs and technology investment.** The paper documents reductions in employment and capital expenditures following the EPS-miss incentive, including lower investment in productivity-augmenting technologies (e.g., machinery and information technology).

- (iv) **Internal reallocation is inefficient: cuts are not targeted to low-productivity units.** Using plant-level data, the paper shows that when resources shrink, firms do not disproportionately cut or exit their least productive plants. Instead, they reduce inputs broadly across plants, including relatively productive plants, which helps explain the measured TFP decline.
- (v) **Frictions amplify the real costs.** The negative productivity effects are stronger when firms face constraints that impede efficient adjustment and reallocation, including labor frictions, financing constraints, short-horizon governance environments, and stronger short-term CEO incentives.